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MILITARY SPECIFICATION

LAST, FOOTWEAR FWU-5/P

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirement for plastic shoe lasts used in the manufacture of quick-donning, lined and insulated flight boots, type FWU-3/P. The lasts are designated as type FWU-5/P plastic.

1.2 Classification. The lasts shall be of one type and in the following sizes and widths, as specified (see 6.2).

SCHEDULE OF SIZES AND WIDTHS

SIZE	WIDTH							
	AA	A	B	C	D	E	EE	EEE
5	X	X	X	X	X	X	X	X
5 1/2	X	X	X	X	X	X	X	X
6	X	X	X	X	X	X	X	X
6 1/2	X	X	X	X	X	X	X	X
7	X	X	X	X	X	X	X	X
7 1/2	X	X	X	X	X	X	X	X
8	X	X	X	X	X	X	X	X
8 1/2	X	X	X	X	X	X	X	X
9	X	X	X	X	X	X	X	X
9 1/2	X	X	X	X	X	X	X	X
10	X	X	X	X	X	X	X	X
10 1/2	X	X	X	X	X	X	X	X
11	X	X	X	X	X	X	X	X
11 1/2	X	X	X	X	X	X	X	X
12	X	X	X	X	X	X	X	X
12 1/2	X	X	X	X	X	X	X	X
13	X	X	X	X	X	X	X	X
13 1/2	X	X	X	X	X	X	X	X

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8335

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* 1.2.1 Part number. Specification part number for items described in this specification will be formulated as shown in section 6.5.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Shipping, Fiberboard.
PPP-B-601	Box, Wood, Cleated - Plywood.

MILITARY

MIL-P-116	Preservation, Methods of.

STANDARDS

FEDERAL

FED-STD-151	Metals; Test Methods.
FED-STD-406	Plastics: Methods of Testing.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D1248-69	Polyethylene Molding and extrusion Materials.
ASTM D-3951	Packaging, Commercial.

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(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article. When specified (see 6.2), before production is commenced, two finished pairs of lasts (one pair from sizes 8 through 10 1/2 and one pair from sizes 11 and over) in any width, shall be submitted to or made ready for the contracting officer or his authorized representative for inspection as specified in 4.3. The approval of the preproduction sample authorizes the commencement of production but does not relieve the contractor of responsibility for compliance with applicable provisions of this specification. The preproduction sample shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production item.

3.3 Material.

* 3.3.1 Polyethylene blocks. The lasts shall be made from polyethelene blocks. The polyethylene blocks shall be made from solid high density polyethylene without fillers, conforming to type III (density 0.941 to 0.965), grade 4 or 5 (melt index 1.9 or less) of ASTM D-1248. The color shall be the standard color used by the contractor, either class A uncolored or class B colored. The blocks shall be made from virgin material, or a mixture of virgin material and reground scrap material of the same composition produced by regrinding clean unburned scrap produced in the fabrication of last blocks, finished lasts and imperfect lasts. Scrap containing dirt or other foreign material shall not be used.

3.3.2 Metal hinges. The metal hinges shall be commercial type, zinc chromated steel to resist moisture and shall be mechanically peened (see 3.5.1).

* 3.3.3 Hinge pins. The hinge pins shall be made of steel wire and shall be heat treated to a hardness value of 45 to 50 Rockwell C scale when tested, as specified in 4.4.1. The pins shall be not less than 15/64 inch nor more than 19/64 inch in diameter. The pins shall be countersunk 1/4 ±1/8 inch under the surface of the sides of the lasts. The pins shall be zinc chromed steel to resist moisture and shall be mechanically peened.

3.3.4 Thimbles. The thimbles shall be manufactured from steel sheet, bar, or tubing stock and shall be heat treated to a hardness value of 55 to 88 Rockwell B scale when tested as specified in 4.4.1. The thimbles may be of the drawn, split, bored, or swaged tubing type. All thimbles shall be full

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flanged with a closed or partially closed bottom, and shall be the lock-in type. Dimensions for last thimbles shall correspond to the following when tested as specified in 4.4.1.

Inside diameter	1/2 ±1/64 inch
Overall length	1 5/8 ±1/16 inches
Depth of hole	1 1/2 ±1/16 inches
Wall thickness	0.0475 to 0.095 inch
Flange diameter	3/4 ±1/16 inch
Flange thickness	75 to 100 percent of wall thickness

3.3.5 Heelplates. The heelplates shall be not less than 3 1/8 inches long on size 8-D, shall measure proportionate to normal grade on other sizes, and shall have a 1/2 inch hole in the center for insole tacking and five countersunk nail holes as specified in 3.5.3. The heelplates shall be made from either hot galvanized or electrogalvanized sheet steel, 0.050 ±0.005 inch thick. The heelplate shall be heat treated to a hardness value of 48 to 67 Rockwell B scale when tested as specified in 4.4.1.

* 3.3.6 Plastic plugs. The plugs shall be of the same material as the body of the lasts or other plastic material having Shore Durometer hardness value of not less than 60 D-scale when tested as specified in 4.4.1. The plugs shall be 1/2 ±1/32 inch in diameter and 3/8 ±1/16 inch long when tested as specified in 4.4.1. The heel plug material shall be a force fit in the drilled tack holes and shall show no evidence of splitting when tested as specified in 4.4.1.

* 3.3.7 Heelplate attaching nails. The nails for attaching the heelplate on the last shall be 15 or 16 gage, 5/8 to 3/4 inch long, barbed, iron wire nails with a 1/8 ± 1/32 inch diameter flat heads and ringed chisel points and shall not be cement coated finish when tested as specified in 4.4.1.

* 3.3.8 Ink. Ink for marking lasts shall be black, shall not blur, and shall have sufficient permanence to meet the requirements specified herein.

3.4 Models, patterns and templates.

* 3.4.1 Turning models. Turning models of each of the sizes and widths required will be furnished by the Government (see 6.3). Turning models shall be in whole sizes 8 and 10 and in widths AA, A, B, C, E, EE, and EEE with the final measurements for the last of that size plainly marked on each model. Only two sizes shall be turned from any model up to size 12, namely, the model size and one-half size up. No last shall be turned from a model of a different width. Sizes 12 shall be turned from the size 12 models.

3.4.2 Replacement of turning models. Models worn to the point of unserviceability in the production of properly graded lasts will be replaced by the Government. All models shall be returned to the Government at the close of the contract. Worn models that need replacement shall be marked "unserviceable."

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3.4.3 Last bottom patterns. Necessary turning bottom patterns and finished bottom patterns will be loaned to the contractor by the Government (see 6.3).

3.4.4 Templates. Templates for use in checking the lasts will be loaned to the contractor by the Government (see 6.3).

3.5 Construction.

* 3.5.1 Hinge cutting, slotting and assembly. The hinge V-cut and circle sawcut shall correspond to the hinge pin holes so that the last, with a correctly inserted hinge, will open and close in a satisfactory manner. All inside sharp corners shall have a slight radius or shall be drilled or run over a hot wire to eliminate stress risers.

* 3.5.2 Thimble hole. The thimble hole shall be bored to such a size that when the thimble is inserted in the last, it shall be locked into the plastic. The flange of the thimble shall seat on top of the last. Thimbles shall be countersunk on lasts. On sizes 7 1/2 through 12 (all widths), the thimble hole shall be centered 1 3/8 and 1 5/8 ±1/16 inches, respectively, from the back of the last top.

* 3.5.3 Attaching heel plates. The heelplate on the finished last shall be flush with, but not overlaying, the outside edge of the last. A narrow margin of plastic, having a maximum width of not more than 1/16 inch in the curved area beginning 3/8 inch from the breast line of the plate, is acceptable. The margin of plastic in the area 3/8 inch from the breast line on each side of the plate is not limited. The breast of the heelplate shall butt up to the plastic. The heelplates shall be attached with five nails specified in 3.3.7. The position of the heel nails shall be according to industry's practice, with countersunk nail holes punched not less than 3/8 nor more than 5/8 inch from the edge of the plate, except that the two front nail holes shall be not less than 3/16 nor more than 3/8 inch from the breast of the heelplate. The side nails shall be located equidistant between the breast nails and the back nail. The heelplates shall be punched with a 1/2 ±1/32 inch tack hole located approximately in the center of the heelplate, measuring both from the breast to the end of the heelplate and from side to side. The heelplates shall be cupped to conform to the contour of the heel seat of the last.

* 3.5.4 Plugging tack holes. Tack hole plugging for the lasts shall be done with heel plug material as specified in 3.3.6. The plug shall be inserted so that it entirely fills the 1/2 ±1/32 inch hole which is drilled through the heelplate to a depth of not less than 3/8 inch nor more than 1/2 inch. After the plug has been inserted, it shall be filed, shaved, or ground even with the heelplate.

3.5.5 Heel and toe making. The heel and toe shall be shaped in accordance with the heel and toe profile templates.

3.6 Measurements and grading.

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3.6.1 Measurements. The contractor shall check the turning models but shall not make any modifications except for slight adjustments to fit the turning model bottom to the Government-furnished paper last bottom patterns. The finished last shall conform to the measurements specified in table I. Variations in the girth measurements for the ball, the waist, and the instep shall not exceed 1/16 inch. Finished last bottom shall conform to the measurements of the finished last bottom patterns furnished by the Government, with a tolerance of plus or minus 1/32 inch in the length and plus or minus 1/48 inch in the width.

TABLE I. Finished measurements (inches).

Size	Width	Ball	Waist	Instep	Stick
8	AA	8 3/8	8 5/16	8 5/8	8 1/8
8	A	8 9/16	8 1/2	8 13/16	8 1/8
8	B	8 3/4	8 11/16	9	8 1/8
8	C	8 15/16	8 7/8	9 3/16	8 1/8
8	D	9 3/16	9 1/8	9 7/16	8 1/4
8	E	9 7/16	9 3/8	9 11/16	8 3/8
8	EE	9 5/8	9 9/16	9 7/8	8 1/2
8	EEE	9 13/16	9 3/4	9 1/16	8 5/8

3.6.2 Grading. The bottom of the lasts shall grade 1/16 inch between widths. The length of the bottoms shall grade up and down from the "E" model as follows: EE model, 1/24 inch longer; EEE model, 2/24 inch longer, D model, 1/24 inch shorter; C model, 2/24 inch shorter; B model, 2/24 inch shorter; A model, 2/24 inch shorter and AA model 2/24 inch shorter.

* 3.7 Marking.

* 3.7.1 Size and width of stamping. The heel part of the saw cut shall be clearly and legibly rubber-stamped with the correct size of the last. The stamping shall be placed on the left side of the hinge cut and shall start not more than 3/16 inch from the bottom saw cut. The correct alphabetic width (AA, A, B, C, D, E, EE, or EEE) of the last shall be clearly and legibly rubber-stamped on the heel part of the saw cut and shall be placed so that the bottoms of the letters are within 1/16 inch of the top of last. The characters used for saw-cut stamping shall be not less than 1/2 inch nor more than 3/4 inch in height and not less than 1/4 inch nor more than 3/8 inch in width. When viewed from directly above the thimble, it shall be possible to simultaneously read all saw-cut stamping. In addition to the stamping on the saw-cut, the size and width designations shall be plainly rubber-stamped on the outside of the last, just ahead of the saw-cut, 1 1/2 inches ±1/2 inch from the top of the cone, to read horizontally. The characters used for this stamping shall be on a line with the size designation. The size and width shall be clearly and legibly die stamped (incised) on the forepart of the last in the tip line area, for example: "10 1/2 AA."

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3.7.2 Marking for identification. In accordance with MIL-STD-130 and using the same size of figures specified in 3.7.1, an identifying inscription shall be marked (incised) with a die stamp on the outside surface of the heel part of each last using not less than 3/8 inch nor more than 1/2 inch characters. The inscription shall be as follows:

TYPE NO. FWU-5/P
U.S.

In addition, the name or symbol of the manufacturer shall be die stamped on the inside surface of the heel part of each last.

3.8 Finish.

* 3.8.1 Finish on last. The last shall be clean, free of plastic hairs, strings, flash, or sprues, and shall have no prominent turning gouges of any kind on the surface. All sharp edges in V-cut or saw-cut shall be given a noticeable radius.

* 3.8.1.1 Resistance to acetone. The marking and writing on finished last shall show no change in appearance when tested as specified in 4.5.2.

* 3.8.1.2 Impact testing. The finished last shall show no chipping or cracking, nor be deformed so as to be unserviceable when tested by impacting with the eight pound ball as specified in 4.5.3.

* 3.8.1.3 Hinge performance. The finished last shall show no cracking, breaking, chipping or other defect which would make the last unserviceable when the last is broken and reclosed 100 times as specified in 4.5.4.

* 3.8.1.4 Tack resistance. The finished last shall show no splitting, chipping, cracking or other defects which would render the last unserviceable when tested with insole tacks as specified in 4.5.5.

* 3.8.1.5 Appearance of last. Depressions, bumps or holes of any kind arising from improper turning or interior voids in the block which appear at last surface shall be cause for rejection.

3.9 Workmanship. The finished lasts shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When required (see 6.2), the preproduction sample submitted in accordance with 3.2 shall be inspected as specified in 4.4.3.1.1 and 4.4.3.1.2 for compliance with design, construction, workmanship and dimensional requirements.

* 4.4. Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition, testing shall be performed on components and materials listed in table II. All test reports shall contain the individual values utilized in expressing the final result. All requirements are applicable to the sample unit. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or services for which in-process inspection has indicated nonconformance.

* 4.4.3 End item inspection.

* 4.4.3.1 End item examination. The defects found during examinations shall be classified in accordance with 4.4.3.1.1 and 4.4.3.1.2. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in 4.4.3.1.3. The sample unit shall be one completely fabricated last. The lot size shall be expressed in units of one last.

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TABLE II Testing of Components

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample Unit
				-	-	-	-
Plastic blocks	Mat'l identification	3.3.1	1/	-	-	-	-
Hinge metal	Mat'l identification	3.3.2	1/	-	-	-	-
	Zinc chromate	3.3.2	1/	-	-	-	-
Hinge pins (1 gross)	Mat'l identification	3.3.3	1/	-	-	-	-
	Hardness value	3.3.3	Method 243 2/	3	-	Avg. Hardness No.	1 pin
	Diameter	3.3.3	1/	-	-	-	-
	Zinc chomate	3.3.3	1/	-	-	-	-
Thimble (1 gross)	Mat'l identification	3.3.4	1/	-	-	-	-
	Hardness value		Method 243 2/	3	-	Avg. Hardness No.	
	Inside diameter	3.3.4	Gage	1	X	-	
	Overall length	3.3.4	Gage	1	X	-	
	Depth of hole	3.3.4	Gage	1	X	-	1 thimble
	Wall thickness	3.3.4	Gage	1	X	-	
	Flange diameter	3.3.4	Gage	1	X	-	
	Flange thickness	3.3.4	Gage	1	X	-	
Heelplate (1 gross)	Mat'l identification	3.3.5	1/	-	-	-	-
	Hardness value	3.3.5	Method 243 2/	3	-	Avg. Hardness No.	1 heelplate
	Thickness	3.3.5	1/	-	-	-	-
	Galvanized	3.3.5	1/	-	-	-	-

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TABLE II Testing of Components (Cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test Method	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
* Plastic plug, (1 gross)	Mat'l identification	3.3.6	1/	-	-	-	-
	Length and diameter	3.3.6	Gage	2	X	-	1 plug
	Hardness value	3.3.6	Method 1082 3/	3	-	Avg. Hardness No.	1 plug
	Resistance to splitting	3.3.6	4.4.1	2	X	-	2 plugs
* Nails and wire (1 lb.)	Mat'l identification	3.3.7	1/	-	-	-	-
	Finish	3.3.7	1/	-	-	-	-
	Length, head diameter and gage	3.3.7	Gage	2	X	-	2 nails or wires

1/ The supplier shall submit a certificate of compliance for this characteristic.

2/ Refers to FED-STD-151.

3/ Refers to FED-STD-406.

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* 4.4.3.1.1 Visual examination.

Examine	Defect
Finish:	Contains plastic hairs, strings, flash or sprues. Depression, bump, hole, or void. Prominent turning gouge on surface.
Construction and workmanship	Any surface of last that will come in contact with the shoe, not smoothly finished, i.e., any depression, protrusion, or rough area. Any part misplaced or out of alignment. Heel end and forepart of last not joined flush, i.e., any step or ridge over 1/64 inch in the outside joint area that will come in contact with the shoe. Any part not a good flush fit where required, i.e., the breast of heelplate to plastic. Any part loose, i.e., heelplate plug is not tight. Any part not recessed in plastic where required, or not recessed to the specified dimension. Any operation omitted or not properly performed. Any functioning part that will not operate as required. Any part missing or damaged. Any nail missing, head bent over, or protruding. More than one nail misplaced. Margin of plastic along back and side edges of heelplate beginning 3/8 inch from breast line in no more than 1/16 inch.
Marking	Omitted, incorrect, illegible, incomplete, not accomplished in specified manner, size of characters not as specified, not in proper location.

4.4.3.1.2 Dimensional examination. The last shall be examined for defects in dimensions. Any dimension that is not within the specified tolerance shall be classified as a defect.

4.4.3.1.3 Inspection level and AQL. The inspection levels and the AQLs expressed in defects per hundred units shall be as follows:

Examination	Inspection level	AQL
Defects applicable to 4.4.3.1.1	II	4.0
Defects applicable to 4.4.3.1.2	S-3	4.0

4.4.4 Packaging inspection. An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping bag fully prepared for delivery except that it need not be sealed. Defects of closure listed below shall be examined on shipping bags fully prepared for delivery. The lot size shall be the number of shipping bags in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component not as specified. Any component missing. Any component damaged, affecting serviceability.
Workmanship	Seams and stitching of bags not as specified. Inadequate application of components such as: Closure of bags not as specified or incomplete.
Content	Number of pairs of lasts per bag is more or less than required.

* 4.4.5 End item testing. The end product shall be tested for the characteristics indicated in table IV. All test reports shall contain the individual values utilized in expressing the final result. For each test characteristic, one test determination shall be made and the results shall be reported as pass or fail. The requirements are applicable to the sample unit for all test characteristics. The sample unit shall be four lasts. The lot size shall be expressed in units of one last. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. End item tests.

End item	Characteristic	Requirement paragraph	Test method
* Finished last	Resistance to acetone	3.8.1.1	4.5.2
	Resistance to impact	3.8.1.2	4.5.3
	Hinge performance	3.8.1.3	4.5.4
	Tack resistance	3.8.1.4	4.5.5

* 4.5 Methods of inspection.

4.5.1 Test for composition plastic heel plug. The plastic heel plug in the finished last shall be subjected to the insertion of a 2 1/2 ounce shoe tack for at least 1/4 inch. The last shall be examined for evidence of the plastic heel plug splitting.

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4.5.2 Test for resistance to acetone. The finished last shall be wet in a test area including marking with acetone for 5 minutes (i.e., using soaked cotton wad). The finish shall be tested by attempting to scrape through the test area with the thumbnail. The last shall be examined for softening of the finish, change in marking and change in appearance from adjacent finish.

* 4.5.3 Impact on plastic lasts. The finished last shall be tested by dropping an 8-pound solid iron or steel ball so as to strike the last in the following spots: (The ball shall be dropped from a height of two feet, measured between the bottom of the ball and the point of impact on the last):

a. With the last resting on its side, the ball shall strike over the rear hinge pin hole and at the edge of the ball line midway between the hinge and the toe.

b. With the last resting on the heel end, the ball shall strike the last on the tip of the toe.

Tests shall be repeated three times in each spot. In the (a) position repeat on both the outside and inside of the last. The last shall be inspected for breaking, splitting, cracking or other defects rendering the last unserviceable, other than directly at the impact area.

NOTE: The last may conveniently be held by a large C-clamp between the thimble and the heel plate and shall be placed on a concrete floor.

* 4.5.4 Test of hinge. The last shall be placed on a regular last spindle and shall be broken and reclosed 100 times. This may be done by hand or by an appropriate hydraulic or mechanical mechanism. Examination should be made for defects developed by this test.

* 4.5.5 Test for resistance to tacks. The finished last shall be tested by driving 2 1/2 ounce insole tacks into the bottom surface of the last, 1/4 inch in from the edge, around the periphery from inside to outside ball area. The tacks shall be driven 1/4 inch deep and shall be spaced 1/2 inch apart.

* 5. PACKAGING

* 5.1 Packaging. Packaging shall be level A, C or Commercial as specified (see 6.2).

* 5.1.1 Level A. Unless otherwise specified (see 6.2), each last shall be individually preserved and packaged per method III, in accordance with MIL-P-116, and shall be placed in a container conforming to PPP-B-636, type CF, Class Domestic.

* 5.1.2 Level C. The level C preservation for lasts shall conform to the MIL-STD-794 requirements for this level.

* 5.1.3 Commercial. Lasts, shall be packaged in accordance with the applicable requirements of ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

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* 5.2.1 Level A. Lasts, when preserved and packaged as specified in 5.1.1, shall be packed in overseas type containers conforming to PPP-B-601. Where practical, the exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the container specification.

* 5.2.2 Level B. Lasts, preserved and packaged as specified in 5.1.1, shall be packed in weather resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and size, shall be limited to the special requirements table of the specification. Where practicable, the exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the container specification.

* 5.2.3 Level C. The level C packing for lasts shall conform to the MIL-STD-794 requirements for this level.

* 5.2.4 Commercial. The preserved lasts shall be packed in accordance with the applicable requirements of ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

* 5.4 Palletization. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet used shall first be approved by contracting officer.

6. NOTES

6.1 Intended use. The lasts covered by this specification are intended to be used as a base over which to manufacture a single design and quality of insulated, quick-donning, flight boots for male members of the United States Air Force. The boot is designated type FWU-3/P and is covered by MIL-B-27486. * Specification MIL-B-83227, boots, flyer's, FWU-8/P uses lasts covered by this specification.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Types, sizes and widths required (see 1.2).

c. First article. When a preproduction sample is required (see 3.2 and 4.2).

d. Selection of applicable level of packing (see 5.1).

e. When palletization is required (see 5.4).

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6.3 Government-furnished property. The following Government-furnished property, in the sizes necessary for performance of the contract, will be loaned by the Government to the contractor for the life of the contract:

a. Last turning models (see 3.4.1).

b. Bottom patterns, turning (see 3.4.3).

c. Bottom patterns, finishing (see 3.4.4).

d. Heel curve templates, finished (3.5.5).

6.4 Commercial designation. The commercial designation of this item is known as "Chippewa Shoe Co., Style 024."

* 6.5 Definitive specification part number. The specification part number is a definitive part number which will be formulated to identify each item covered by this specification. The part number will be formulated by selecting from the requirement options available in this specification as follows:

Definitive specification part number: M27885 - X - X - X

Military specification number

Type designator (see 6.5.1.1)

Size designator (see 6.5.1.2)

Width designator (see 6.5.1.3)

* 6.5.1 Type designator. A one position field used to designate type.

* 6.5.1.1 Size designator. A one position field used to designate size. (A-5, B-5 1/2, C-6, D-6 1/2, E-7, F-7 1/2, G-8, H-8 1/2, I-9, J-9 1/2, K-10, L-10 1/2, M-11, N-11 1/2, O-12, P-12 1/2, Q-13, R-13 1/2).

* 6.5.1.2 Width designator. A one position field used to designate width. (1-AA, 2-A, 3-B, 4-C, 5-D, 6-E. 7-EE, 8-EEE).

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 99

Review activity:
DLA - CT

User activity:
Air Force - 45

Preparing activity:
Air Force - 82

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-27885C	LAST, FOOTWEAR FWU-5/P

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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